

MuTau channel: status and next steps

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Current results — $\mu\tau_h$ channel

- Signal: $pp \rightarrow p(\gamma\gamma \rightarrow \tau^+\tau^-)p$, $\sqrt{s} = 13$ TeV, 54.9 fb^{-1} (2018)
- BDT-based analysis (Scan A: AdaBoost, $N = 600$, $d = 2$)
- $\sigma_{\text{SM}} = 4.7 \text{ fb}$, per-arm proton selection

Quantity	Value
Expected -2σ	$r < 37.1$
Expected -1σ	$r < 56.1$
Expected (median)	$r < 95.0$
Expected $+1\sigma$	$r < 176.0$
Expected $+2\sigma$	$r < 288.0$
Observed	$r < 281.6$
Observed significance	2.41σ

Thesis outline

Ch.	Title	Status
1	Theory (paper reviews)	draft
2	Introduction (SM, τ , g -2, CEP)	in progress
3	Experimental apparatus (CMS + PPS)	draft
4	Preliminary analysis	draft
5	Samples and object reconstruction	in progress
6	Analysis and results ($\mu\tau_h$, BDT, Combine)	in progress